Cohort-Anchored Infant EEG State-Space Geometry Tracks Age in Early Infancy

Child-clustered coordinate trajectories, with within-child local-dynamics robustness checks

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Abstract We examined age-associated change in EEG-derived state-space coordinates in 70 mixed rest/reaching recordings from 22 typically developing infants (38-203 days; 20 with repeated visits) in the public Figshare cohort, after masking artifact-flagged (BAD_-interval) segments before epoching (5,265 of 6,929 candidate epochs retained). A frozen cohort-level IQR anchor placed every recording in a common 3D/9D coordinate chart. Child-clustered Gaussian GEE models with a cubic age spline showed age association at all 12 cohort-coordinate endpoints after FDR correction, driven by a within-child rather than between-child effect (within-child $p < 0.011$ at every endpoint; no between-child term FDR-significant). A more conservative child-cluster permutation check supported only 6 of 12 endpoints after its own FDR correction; the mobility and entropy-related coordinates were least robust, the latter also failing a direct-age sensitivity check. Two independent local-dynamics checks - within-child MNJ/reachability, and a cohort-pooled, leave-one-child-out reachability estimate - showed no FDR-supported age effect in either primary family; one direct-age-subset endpoint (cohort-pooled rotational proxy, raw $p=0.040$, not FDR-supported) remained nominally significant, while a separately noted nominal within-child hint present before artifact masking disappeared entirely afterward, consistent with a motion-artifact origin for that earlier hint. As a descriptive comparator, 34 conventional qEEG measures (relative band power, band ratios, spectral peaks, entropy/complexity, and inter-channel coherence) extracted from the same 5,265 epochs showed a materially stronger and, for a subset, more robust age association than either local-dynamics family: 24 of 34 were FDR-supported at the cohort level, driven overwhelmingly by a within-child rather than between-child effect, and 3 endpoints (the beta/alpha power ratio, relative beta power, relative gamma power) survived both a conservative cluster-permutation check of the cohort model and an independent child-sign-permutation check of the within-child slope. These results show a within-child coordinate association that is robust for roughly half, but not all, endpoints under conservative testing, alongside well-established classical qEEG maturation markers detected in the same cohort, without an established corresponding change in local flow geometry.

Author Summary

We tracked EEG-derived state-space coordinates across infant recordings using a shared, cohort-level coordinate chart, after masking motion/artifact-contaminated segments. Coordinates changed with age within individual children, not just between them, but a stricter permutation-based check supported only half of the endpoints once corrected for multiple testing, flagging two (mobility, entropy-related) as fragile. Two checks of local flow-geometry summaries - within-child and cohort-pooled - found no FDR-supported change, though one cohort-pooled direct-age-subset endpoint stayed nominally (uncorrected) significant; a separate small hint seen before artifact masking disappeared once artifact-contaminated segments were removed. As a sanity check, classical EEG maturation markers (band power, band ratios, entropy, coherence) computed from the same recordings did show age-related change; the beta/alpha ratio and relative beta/gamma power survived both strict robustness checks, showing this cohort and pipeline can detect established developmental EEG signal even where the newer local-dynamics summaries did not.

1. Introduction

Longitudinal infant EEG can characterize developmental change, but irregular recording schedules make child-level inference and calibration especially important. We analyse a public Figshare infant EEG cohort (Xiao et al.,

2017) using exported EEG-derived state-space summaries. “State-space geometry” here denotes the geometry of a versioned feature chart, not a directly observed neural state or latent neural manifold.

1.1. Related evidence and dataset boundary

The Figshare release analysed here has also supported separate studies of infant power and coherence, periodic/aperiodic activity, and early cognition and language associations (Bradley et al., 2024; Schaworonkow & Voytek, 2021; Xiao et al., 2018). Those papers provide same-corpus provenance and complementary measurements, not independent replication of the present coordinate analysis.

A related but distinct resource is OpenNeuro `ds004577`, which contains 130 resting EEG recordings from 103 typically developing infants during the first year of life, primarily during physiological sleep (Harmony et al., 2023). Its accompanying dataset article describes 15 days–12.5 months, 200-Hz recordings, and sleep-stage-II sampling (Harmony et al., 2024). Earlier work from the same research program, not necessarily the identical OpenNeuro sample, reported normative qEEG trajectories, source-space spectral development, and age-related effective-connectivity patterns (Bosch-Bayard et al., 2012; 2022; Otero et al., 2011). The present study uses the Figshare release only; `ds004577` was not pooled, used as an anchor, or treated as a replication sample.

Independent infant EEG cohorts likewise report developmental change in spectral composition and connectivity, including aperiodic/periodic trajectories across 2–44 months, relative-power decomposition, and network-level maturation (Falivene et al., 2024; Hu et al., 2022; Rico-Picó et al., 2023; Wilkinson et al., 2024). Complexity and inter-individual variability are also active targets of infant EEG research, including longitudinal multiscale entropy in preterm infants and cross-sectional comparisons of power, connectivity, entropy, and fractal dimension (Davoudi et al., 2023; Pelc et al., 2022). These findings motivate the conventional-qEEG comparator and the cautious separation of chart position from local-flow geometry in this article.

Core question

Do common-scale, cohort-anchored MNPS coordinates show age-related change across this cohort, and does that change also appear as a within-child local-dynamics (MNJ/reachability) effect?

2. Theory

Let x_t denote the three-dimensional EEG-derived MNPS chart and $f(x,t)$ its estimated local flow. The MNJ is the local linearisation

$$J(x,t) = \partial \frac{f}{\partial} x. \tag{1}$$

We summarize its magnitude, trace, rotation, and spectral radius, and derive short-horizon reachability summaries from the associated local linear model. The identical construction is applied unchanged in the nine-dimensional MNPS chart (9D $x_t \in \mathbb{R}^9$), used both for the primary cohort-coordinate family’s nine 9D endpoints (Results R2) and as a secondary 9D MNJ sensitivity family (Results R4, Supplement S2.6); we do not repeat the derivation for that case. These operational statistics describe the exported chart and do not uniquely identify a biological mechanism.

3. Methods

3.1. Dataset and export contract

The Figshare source is a longitudinal typical-development dataset (Xiao et al., 2017). The analysed export contains 70 H5 recordings from 22 children, with visits 1-6 and ages 38-203 days. Sixty-nine geometry contracts were `ok` and one was `adjusted` under the standard invalidity policy; all were retained and all 70 visits provided 3D and 9D MNJs. The upstream export used 256-Hz EEG, 0.3-30-Hz filtering, 8-s windows with 4-s steps. This analysis uses an artifact-masked re-export: the upstream ingest pipeline’s BAD_-interval annotations (715 artifact intervals, 2,143.7 s total, from the publicly released BAD_ files) are masked at the sample level before epoching, so any 8-s epoch overlapping a flagged interval is dropped rather than bridged; this reduced the usable epoch pool from 6,929 to 5,265 epochs (median 78.5 per session, range 20-109), down from a prior median of 98 (range 30-148) in the pre-masking export. The frozen subject-balanced IQR anchor `figshare_infant_eeg_global_cohort_all_visits_iqr_v2_1` was refit on this artifact-masked export (anchor hash `45bbfa21...c301a89e`) and provides both subject- and cohort-anchored layers. Rest/reaching condition labels are not yet available upstream; all recordings are analysed as mixed rest/reaching sessions, and BAD_ annotations are used only for artifact QC, not condition labelling.

3.2. Analysis design

The visit, not the epoch, was the inferential unit. The primary analysis is the cohort-anchored coordinate model: a Gaussian GEE with exchangeable child clustering and a 3-df cubic age spline fit to 3D and 9D cohort-coordinate medians, with FDR applied across the 12-coordinate family (Benjamini & Hochberg, 1995). Two robustness checks were added to this primary model: (1) a within/between-child decomposition that replaces the raw age spline with a linear child-mean ("between") age term plus a cubic spline of centered ("within") age, testing each component separately; and (2) a child-cluster row-permutation check of the omnibus Wald statistic (age values permuted across all visit-rows, the identical GEE refit 500 times per endpoint), which does not rely on asymptotic sandwich-covariance theory and is therefore a useful check given only 22 clusters.

As a robustness complement to the coordinate result, two independent local-dynamics analyses assessed whether the same age association is also present as a change in local flow geometry. The first is a within-child analysis of each session's own MNJ Frobenius norm, rotation norm, trace, spectral radius, and reachability log-volume, effective dimension, anisotropy, directional entropy, and rotational proxy (each session's reachability estimated only from that session's own epochs); it removes each child's mean age and endpoint value, then uses child-cluster bootstrap intervals and child-sign-permutation p-values. Twenty children (68 endpoint-complete visits) contributed to this within-child robustness check; full results, including a direct-age-only sensitivity subset, are reported in Supplement S2. The second is a cohort-pooled reachability analysis: at each visit's own cohort-anchored 3D coordinate, a local linear flow model is fit from the k=12 nearest epoch-to-epoch transitions drawn from every **other** child's epochs (leave-one-child-out), using the same ridge-regression and reachability-propagation machinery as the within-session estimate; the resulting per-visit reachability summaries are tested with the identical child-cluster estimator used for the within-child family.

3.3. Conventional-EEG comparator family

As a descriptive comparator to the cohort-coordinate result, we also tested 34 conventional quantitative-EEG ("qEEG") endpoints extracted from the same 5,265 artifact-masked epochs by the upstream ingest pipeline's `tier1`, `complexity`, and `connectivity` comparator packs: relative band power (delta/theta/alpha/beta/gamma and two infant-specific alpha/theta bands), band-power ratios (including a slowing index), spectral peak/edge frequencies, permutation and spectral entropy, Hjorth complexity, and alpha/theta inter-channel coherence at four channel pairs. Each visit's value is the per-epoch median of the corresponding raw comparator column; the identical cohort-level spline GEE, within/between decomposition, and child-cluster row-permutation check used for the primary cohort-coordinate family (Methods, above) were applied to this family, plus the within-child sign-permutation slope estimator (Methods S1.3-style) used for the local-dynamics families, all with their own, separate BH-FDR correction across the 34 endpoints. This is a separate, purely descriptive robustness/context family; it is not pooled with, and does not alter the FDR correction of, the primary cohort-coordinate family (R2) or either local-dynamics family (R4, R5).

3.4. Boundaries and robustness

Cohort-anchored coordinate positions are interpretable only under this export's frozen anchor and are not pooled numerically with APPLESEED (Puglia et al., 2022). Subject-anchored coordinates remain descriptive for within-child use. Cohort-coordinate robustness restricts the GEE analysis to 40 spreadsheet-direct age records; within-child robustness uses 36 eligible visits from 18 children. Bayley fields are retained for future work but are not analysed as outcomes.

4. Results

4.1. R1: Coverage and QC

The cohort had 70 visits from 22 children; 20 had repeated recordings. Age coverage was uneven and visit number was not a fixed developmental age stratum. All visits had resolved 9D MNJs; one was retained as an `adjusted` geometry contract after artifact masking (down from two in the pre-masking export).

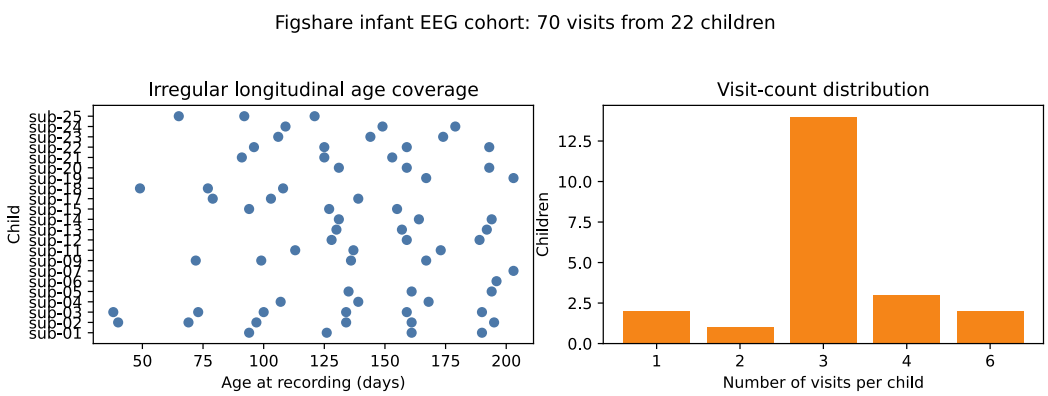


Figure 1: Age coverage and visit density. The irregular schedule motivates within-child age-days rather than visit-number inference.

4.2. R2: Cohort-anchored age trajectories (primary)

All 12 cohort-coordinate endpoints showed an age association in child-clustered spline GEE models after FDR correction. In 3D, deviation was the strongest coordinate result (Wald $\chi^2 = 54.72$, raw p=7.86 × 10⁻12, FDR p=4.72 × 10⁻11); entropy-related coordinate was more modest ($\chi^2 = 15.09$, FDR p=2.09 × 10⁻3); mobility, the weakest 3D result, improved after artifact masking relative to the pre-masking export ($\chi^2 = 11.28$, raw p=0.0103, FDR p=0.0112). The strongest result overall was 9D m_o ($\chi^2 = 155.37$, FDR p=2.19 × 10⁻32); the weakest was 9D d_s ($\chi^2 = 7.89$, raw and FDR p=0.048). These are common-chart position associations, not direct local-flow or neural-state effects.

The within/between decomposition (Methods) shows this is a genuine within-child effect: at every one of the 12 endpoints the within-child spline term is significant (all raw p < 0.011; strongest: 9D m_o, $\chi^2 =$ 337.4, p=7.94 × 10⁻73; weakest: 9D d_s, $\chi^2 = 11.25$, p=0.0105), while no endpoint's between-child linear term is FDR-significant (raw p range 0.04-0.98). The child-cluster row-permutation check is a substantially more conservative, finite-sample-robust test of the same omnibus statistic: after BH-FDR correction of the 12 permutation p-values, 6 of 12 endpoints remained supported - 9D m_o (permutation FDR p=0.024), and 3D deviation, 9D m_e, 9D d_n, and 9D e_e/e_m (permutation FDR p=0.040 each). Six endpoints did not survive: 3D mobility and 3D entropy-related coordinate (permutation FDR p=0.152 each), 9D m_a and 9D e_s (p=0.075 each), 9D d_l (p=0.062), and 9D d_s (p=0.222). We therefore report the cohort-coordinate result at two confidence levels: the asymptotic GEE supports all 12 endpoints and a within/between decomposition confirms the effect is within-child rather than cross-sectional at all 12, but a conservative FDR-corrected cluster-permutation check supports only half (6 of 12). Full per-endpoint decomposition and permutation tables (including the FDR-corrected permutation p-values) are in Supplement S2.

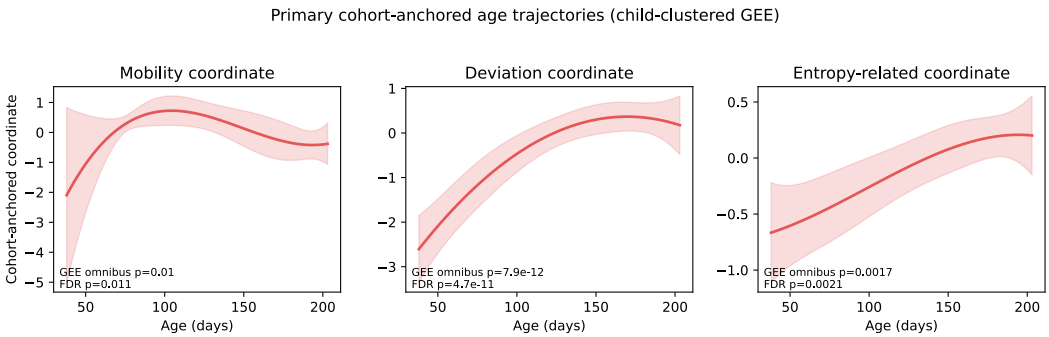


Figure 2: Cohort-anchored 3D coordinate trajectories from child-clustered spline GEE, the primary result of this study. Curves describe common-chart coordinate position, not directly observed neural states.

4.3. R3: Age-source sensitivity of the cohort-coordinate result

In the 40-record spreadsheet-direct age sensitivity, 11 of 12 cohort-coordinate models remained FDR-supported; only the 3D entropy-related coordinate was not (raw/FDR p=0.070). The 3D mobility coordinate was more strongly supported under this sensitivity (FDR p=6.62 × 10⁻4) despite failing R2′s cluster-permutation check, whereas the entropy-related coordinate now fails **both** the direct-age sensitivity and the cluster-permutation check, making it the single least robust endpoint in this analysis. The two checks probe different aspects of

robustness (age-source dependence vs. finite-sample cluster inference) and need not agree, but an endpoint failing both warrants particular caution.

4.4. R4: Within-child local-dynamics robustness check

As a robustness complement to R2-R3, we asked whether the cohort-level age association also appears as a within-child local-dynamics effect estimated from each session's own epochs. None of the nine primary 3D MNJ/reachability endpoints was FDR-supported; the smallest raw p was for the MNJ trace (standardized slope −0.22, raw p=0.054, FDR sensitivity p=0.484), whose child-bootstrap interval excluded zero even though its sign-permutation p was just above the nominal threshold. A secondary 9D MNJ sensitivity family and a direct-age-only within-child subset were also tested, with no FDR-supported endpoint (smallest raw p=0.076 and 0.208, respectively). Notably, two 3D endpoints that were nominally significant (uncorrected) in the direct-age subset of the pre-masking export (`frobenius_norm_median` and `rotation_norm_median`, raw p=0.043-0.050) are no longer nominally significant after artifact masking (raw p=0.277 and 0.208); this pattern is consistent with those earlier nominal hits having been driven by residual motion artifact rather than a developmental effect, though this cannot be confirmed directly and the corrected/masked comparison is descriptive, not a formal before/after test. Full results, figures, and the direct-age within-child sensitivity numbers are in Supplement S2. This null is compatible with, and does not contradict, the R2 cohort-level result: developmental movement across cohort-chart position need not imply a detectable within-child change in the local Jacobian or short-horizon reachability summaries at this sample size.

4.5. R5: Cohort-pooled reachability robustness check

As a second, independent local-dynamics test, we estimated reachability at each visit's own cohort-coordinate position from a local flow model fit exclusively to **other** children's nearby epoch-to-epoch transitions (leave-one-child-out; median 8 distinct contributing children per visit, from a pool of 5,195 transitions). None of the five cohort-pooled reachability endpoints was FDR-supported or nominally significant in the primary test; the smallest raw p was for the rotational-proxy endpoint (standardized slope −0.25, raw p=0.196). The direct-age-only subset showed a single nominal (uncorrected) hit for the same rotational-proxy endpoint (standardized slope −0.47, raw p=0.040, FDR sensitivity p=0.200), not FDR-supported. The pooled neighbor radius was stable across age terciles (median 0.35, 0.29, and 0.33 in the youngest, middle, and oldest thirds of the sample), so these nulls are not attributable to markedly uneven cohort density at different ages.

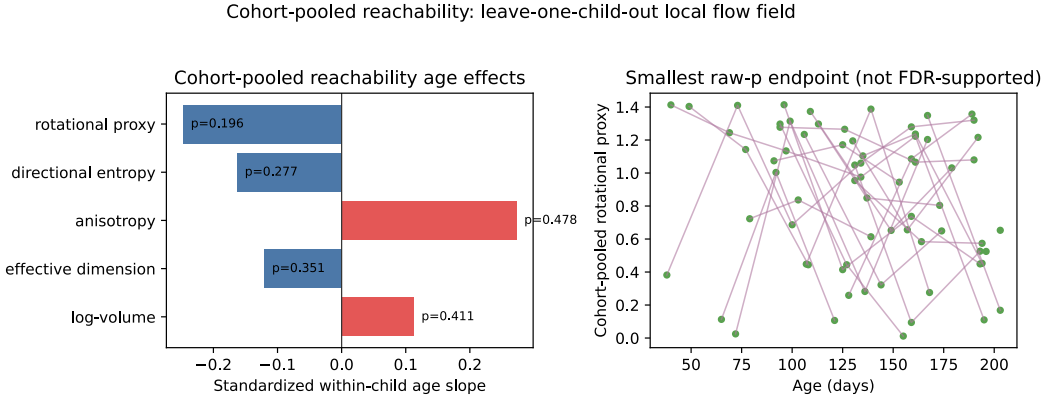


Figure 3: Cohort-pooled reachability age effects (leave-one-child-out local flow field) and the family's own smallest raw-p endpoint (rotational proxy), not FDR-supported.

4.6. R6: Conventional-EEG (qEEG) comparator family

As a descriptive comparator to R2-R5, we tested 34 conventional qEEG endpoints extracted from the identical 5,265 artifact-masked epochs (Methods). Eighteen endpoints (relative band power, band ratios, spectral peaks, entropy/complexity) had 100% session coverage; the remaining 16 (alpha/theta coherence at four channel pairs) had 80-96% coverage, lower at the F8-P4 and central-parietal/frontal-central pairs than at F4-P4, most likely reflecting per-recording channel/segment availability rather than a processing error. In the cohort-level spline GEE, 24 of 34 endpoints were FDR-supported: 14 of 18 non-connectivity endpoints (the four exceptions were the alpha/theta ratio, delta/alpha ratio, slowing index, and theta/alpha ratio) and 10 of 16 connectivity endpoints. The within/between decomposition showed this is overwhelmingly a within-child effect, as in R2: the within-child spline term was FDR-supported at 31 of 34 endpoints (vs. only 1 of 34 for the between-child linear term), including the two most extreme cohort-level effects (alpha frontal-central coherence mean and SD; within-child $\chi^2 = 128.7$ and 31.1, both raw $p < 10^{-}6$, vs. between-child raw p=0.22 and 0.16).

The same conservative child-cluster row-permutation check applied in R2/S2.3 supported only 5 of 34 endpoints after its own FDR correction: alpha frontal-central coherence mean and SD (permutation FDR p=0.023 each), the beta/alpha power ratio (0.023), and relative beta and gamma power (0.027 each). An independent within-child sign-permutation check of the demeaned slope (the same estimator used for R4/R5) supported a materially overlapping but not identical set of 6 endpoints: permutation entropy, the beta/alpha ratio, the slowing index, relative beta and gamma power, and theta frontal-central coherence mean (FDR sensitivity p=0.007-0.044). The intersection comprised three spectral endpoints – the beta/alpha ratio and relative beta/gamma power – and these are therefore the most robust conventional-EEG endpoints in this family. Their age directions are consistent with prior longitudinal work documenting developmental shifts in infant EEG spectral composition and power (Marshall et al., 2002; Schaworonkow & Voytek, 2021; Xiao et al., 2018).

The two alpha frontal-central coherence endpoints (mean and SD) reached the most extreme cohort-level test statistics of the entire study (Wald $\chi^2 = 297$ and 434, raw $p \approx 10^{-}64$ and $10^{-}94$). They survived the cluster-permutation check but not the independent within-child sign-permutation check (within-child FDR-sensitivity p=0.075 for the mean and 0.980 for the SD), so they did not survive both robustness checks. We flag this specific pair for caution: an effect this extreme, in a comparator with only 80% session coverage, is at least as consistent with an age-correlated non-neural confound (e.g. residual signal-to-noise or motion-artifact rate, or cap-fit/head-growth-related electrode contact, both of which plausibly improve with age) as with a genuine coherence-maturation effect, and we cannot distinguish these possibilities with the data and checks available here.

The qEEG comparator citations are contextual rather than independent replication: Xiao et al. (2018) and Schaworonkow and Voytek (2021) analysed the same Figshare corpus, whereas Marshall et al. (2002) provide an independent developmental-EEG reference. The comparator therefore confirms detectability in this pipeline, not external replication.

5. Discussion

The primary result of this reanalysis is that cohort-anchored coordinate positions show age association in a common EEG-derived chart across 70 visits from 22 children, and a within/between-child decomposition shows the effect is driven by within-child movement rather than between-child differences at all 12 endpoints. That said, the strength of evidence depends materially on which robustness check is applied: the asymptotic FDR-corrected GEE supports all 12 endpoints, but a child-cluster row-permutation check - designed specifically because 22 clusters is a known regime for anti-conservative sandwich-covariance inference - supports only 6 of 12 endpoints once its own p-values are FDR-corrected. This is exactly the small-cluster fragility that motivated adding the check, and it substantially tempers the “all 12 endpoints” framing of the asymptotic result: readers should treat 9D m_o as the single most robust result, 3D deviation and 9D $m_e/d_n/e_e/e_m$ as moderately robust, and the remaining six endpoints (most notably 3D mobility and 3D entropy-related coordinate, which also fails the direct-age sensitivity check) as supported only by the asymptotic test.

This reanalysis was itself motivated by an artifact-QC improvement: masking EEG segments flagged as motion/artifact-contaminated by the upstream BAD_-interval pipeline removed roughly a quarter of the epoch pool (6,929 to 5,265 epochs) and slightly **strengthened** the primary cohort-coordinate result (e.g. 3D mobility’s asymptotic FDR p improved from 0.034 to 0.011), while a previously-noted nominal signal in the within-child local-dynamics direct-age sensitivity subset (two 3D endpoints at raw p=0.043-0.050) disappeared entirely (both now raw p > 0.2). Both changes point the same direction: the coordinate-level developmental association is not an artifact of residual movement contamination, and the earlier nominal local-dynamics hint plausibly was.

Two independent local-dynamics checks - one estimated purely from each session’s own epochs (within-child), one estimated from a leave-one-child-out cohort-pooled flow field at each visit’s own coordinate - did not detect an FDR-supported age effect at this sample size. The within-child family showed no nominal signal in any tested subset after artifact masking; the cohort-pooled family was likewise null in its primary form, though its own direct-age-only subset retained one nominal, uncorrected hit (rotational proxy, raw p=0.040; Results R5). Because the cohort-coordinate model pools within- and between-child age information while both local-dynamics checks isolate a different, more local estimand, these are compatible but distinct results, not a contradiction: cohort-chart position moves with age, in a way that is robustly attributable to within-child change, without a correspondingly detectable change in either session-level or cohort-pooled local flow geometry.

The conventional-EEG comparator family (R6) helps interpret both the R2 fragility pattern and the R4/R5 nulls. First, the same child-cluster row-permutation check that reduced R2′s asymptotically-significant 12/12 to a robust 6/12 also reduced R6′s asymptotically-significant 24/34 to a robust 5/34. This supports reading the permutation check’s conservatism as a general property of this 22-cluster design, while recognizing that the two families also differ in endpoint count and that 500 permutations limit the resolution of their permutation p-

values. Second, previously reported qEEG maturation markers – the beta/alpha power ratio and relative beta/gamma power – were detected, in the same within-child-dominant pattern as R2, using the same 5,265 epochs and the same estimators that found no local-dynamics effect in R4/R5 (Marshall et al., 2002; Schaworonkow & Voytek, 2021; Xiao et al., 2018). This weighs against attributing the R4/R5 local-dynamics nulls simply to insufficient signal, artifact contamination, or an underpowered design: this cohort and pipeline can detect age-related EEG change when it is present in a classical spectral/entropy form, so the R4/R5 nulls should be read as evidence that this specific age effect is not (yet, at this sample size) visible in the local Jacobian/reachability construction, not as a generic failure to detect anything.

The main internal contribution is a contract-aware pipeline: it retains session-level inference, freezes a subject-balanced common anchor, masks artifact-flagged segments before epoching, separates chart-position from local-dynamics claims, decomposes within- from between-child age effects, adds a permutation-based robustness check to the asymptotic GEE test, and tests local dynamics both within-child and cohort-pooled. A future age-balanced cohort with an independently fitted or cross-fitted anchor should test whether the R2 coordinate trajectories, and the R2 within/between decomposition, replicate.

6. Limitations

The source combines mixed rest/reaching EEG without event-level labels (BAD_-interval annotations are used only for artifact QC, not condition labelling), lacks stage labels, and has uneven follow-up. Thirty age values are derived from first-visit age and recording date. The cohort anchor was fit using all usable ages, so it is suitable for descriptive common-scale association but is not a held-out normative reference. The spline is exploratory. Only 6 of the 12 cohort-coordinate endpoints survive a conservative, FDR-corrected child-cluster permutation check of the omnibus test, even though all 12 pass the asymptotic FDR-corrected GEE and the within/between decomposition; the 3D mobility and 3D entropy-related coordinates are the least robust (the latter also fails the direct-age sensitivity check) and should be treated as fragile pending replication, and the remaining endpoints supported only by the asymptotic test should be read with corresponding caution. The cohort-pooled reachability estimate (R5) uses a single local fit per visit rather than a within-session tube of estimates, so it is not a strictly larger-sample version of the within-child check (R4) but a distinct estimand with its own assumptions, most notably that leave-one-child-out neighbor density is roughly comparable across ages (checked descriptively, Results R5). Both local-dynamics robustness checks (R4, R5, Supplement S2) are limited by modest child count and irregular sampling and should not be read as evidence of developmental invariance in local dynamics. The comparison between this artifact-masked export and the earlier pre-masking export is descriptive, not a formal statistical test of the effect of artifact masking, since it involves two different, non-independent versions of the same underlying recordings rather than a controlled ablation. The conventional-EEG comparator family (R6) is purely descriptive context, not a primary or confirmatory analysis, and its 34-endpoint FDR family is independent of, and not comparable in family size to, the primary cohort-coordinate family (R2) or either local-dynamics family (R4, R5); it uses a second, separate H5 re-extraction (`results/README.md`) that we verified numerically identical to the primary export on every shared MNPS/MNJ/reachability column but which differs in one respect relevant to R6 itself: the 16 `connectivity` comparator endpoints have only 80-96% session coverage, and the two most extreme R6 effects (frontal-central coherence) are flagged in R6 as plausibly confounded by an age-correlated non-neural factor (signal quality or electrode contact) rather than as an established coherence-maturation effect.

7. Conclusion

In this Figshare infant cohort, cohort-anchored MNPS coordinate positions changed with age across children, and this effect is attributable to within-child movement rather than between-child differences at every endpoint tested. However, only 6 of 12 endpoints remain supported once a conservative, FDR-corrected child-cluster permutation check is applied, and the 3D mobility and entropy-related coordinates are fragile under at least one robustness check each. Two independent local-dynamics robustness checks - within-child and cohort-pooled - showed no FDR-supported effect at this sample size after artifact masking; the within-child family showed no nominal signal at all, while the cohort-pooled family's own direct-age-only subset retained a single nominal, uncorrected hit (rotational proxy), and a separately noted nominal local-dynamics hint present before artifact masking disappeared entirely once artifact-contaminated segments were removed. This is evidence of a within-child developmental association in a common coordinate chart for a subset of endpoints, not a robustly established effect across the full 12-endpoint family, and not yet a confirmed local-dynamics or neural mechanism. A separate, descriptive comparator of 34 conventional qEEG endpoints in the same recordings found a within-child-dominant age association at 24 of 34 endpoints asymptotically and 5 of 34 (beta/alpha ratio, relative beta/gamma power, frontal-central coherence) under a conservative cluster-permutation check, confirming that

established EEG maturation signal is detectable in this cohort with this pipeline, and contextualising, without resolving, the R4/R5 local-dynamics nulls.

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Data Availability Statement

The source data are available through Figshare (Xiao et al., 2017). Analysis wrappers and derived result tables are in this repository under `articles/infant_eeg/`.

Competing Interests

The author declares no competing interests.